

Chapter 8: Acids and Bases

Acids

An acid is a substance that ionises or dissociates in water to produce hydrogen ions.

Acid	Formula	Ions produced in aq
hydrochloric acid	HCl	H^+, Cl^-
nitric acid	HNO_3	$\text{H}^+, \text{NO}_3^-$
sulfuric acid	H_2SO_4	$\text{H}^+, \text{SO}_4^{2-}$
ethanoic acid	CH_3COOH	$\text{H}^+, \text{CH}_3\text{COO}^-$

Properties of Acids

1. Physical Properties of Acids

- Acids have a sour taste.
- Acids produce ions when dissolved in water, and these mobile ions produced act as mobile charge carriers to conduct electricity.
- Acids turn blue litmus paper red.
- Acids turn Universal Indicator red, orange or yellow.
- Acids have pH less than 7.

2. Chemical Properties of Acids

- Acids reacts with Bases (metal oxides and metal hydroxides) to form a salt and water.

Chemical Reaction: acid + base $\rightarrow$ salt + water

Determine the salt produced: (Salts have no charge)

positive ion of metal + negative ion of acid

Example: copper(II) oxide + sulfuric acid $\rightarrow$ copper(II) sulfate



positive ion of metal $\uparrow$ negative ion of acid $\uparrow$

Reaction between acid and base is also known as neutralisation.

b. Acid reacts with reactive metals to form a salt and hydrogen gas.

Chemical Reaction: acid + metal (reactive) $\rightarrow$ salt + hydrogen

Determine the salt produced:

positive ion of metal + negative ion of acid

Test for hydrogen gas

- Place a lighted splint at the mouth of the test tube.
- The lighted splint will extinguish with a 'pop' sound if hydrogen gas is present.

Unreactive metals (do not react with acids)

- Copper, silver, gold

Metals that react explosively with acids

- Sodium, Potassium

c. Acid reacts with carbonates to form a salt, water and carbon dioxide gas.

Chemical Reaction: acid + carbonate $\rightarrow$ salt + water + carbon dioxide gas

Determine the salt produced:

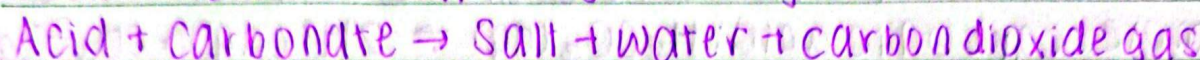
positive ion of carbonate + negative ion of acid

Test for carbon dioxide gas

- Bubble the gas through limewater (aqueous calcium hydroxide)
- White precipitate will be formed in limewater if carbon dioxide gas is present.

Summary

Chemical Reactions of Acids:



What causes acidity?

Pure acid does not behave as an acid unless water is present. Water is essential for the formation of H^+ ions.

- Molecules of acid only dissociate into hydrogen ions in the presence of water, displaying its acidic properties.
- Without water, no hydrogen ions will be produced.

Concentration of Acids

Concentration of acid refers to the number of moles of acid dissolved in each litre or dm^3 of water.
e.g. 0.01 mol/dm^3 is less concentrated than 0.1 mol/dm^3 .

- Concentration of acid can be increased or decreased by adjusting the volume of water or amount of acid used.

Strength of Acids

The strength of an acid refers to the extent of dissociation of the acid molecule into ions when dissolved in water.

Strong Acids

- Strong acids dissociate completely in water to produce hydrogen ions.



- Other strong acids: nitric acid, hydrochloric acid, sulfuric acid

- Properties

→ very low pH (around 1) → very good electrical conductivity

Weak acids

- Weak acids dissociate partially in water to produce a low concentration of hydrogen ions.

- Most of the acid remains as undissociated molecules.

Example: carbonic acid: $\text{H}_2\text{CO}_3(\text{aq}) \rightleftharpoons 2\text{H}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq})$

(Note: $\rightleftharpoons$ means the reaction is reversible)

Other weak acids: carbonic acid, H_2CO_3 ;

ethanoic acid, CH_3COOH ;

sulphurous acid, H_2SO_3

- Poor electrical conductivity

Note

- Strength of an acid $\neq$ concentration of an acid

- The strength of an acid depends on the extent of dissociation while the concentration of an acid depends on the amount of acid dissolved per unit volume.

Basicity of acids

Basicity of acids refers to the number of hydrogen atoms in one molecule of acid that can be replaced by a metal. It refers to the number of H^+ ions produced by one molecule of the acid when dissolved in water.

Monobasic $\rightarrow$ one molecule of acid produces one H^+ ion when dissolved in water

Examples: HCl , HNO_3 , CH_3COOH

Dibasic $\rightarrow$ one molecule of acid produces two H^+ ions when dissolved in water

Examples: H_2SO_4 , H_2SO_3 , H_2CO_3

Tribasic $\rightarrow$ one molecule of acid produces three H^+ ions when dissolved in water

Examples: H_3PO_4

Bases

Bases refer to a class of substances which include all metal oxides and hydroxides.

- Bases react with acids to form salt and water only.
- All bases are insoluble in water except alkalis! (SPCAB)

Examples:

sodium oxide, Na_2O

zinc oxide, ZnO

copper(II) oxide, CuO

Alkalis

Alkalis are soluble bases that ionises in water to form hydroxide ions. (OH^- ions)

___ oxide + water $\rightarrow$ ___ hydroxide (always aq)

(oxides dissolve in water to form hydroxides)

Examples: (SPCAB)

sodium hydroxide, NaOH

potassium hydroxide, KOH

calcium hydroxide, Ca(OH)_2

aqueous ammonia, $\text{NH}_3(\text{aq})$

barium hydroxide, Ba(OH)_2

Strength of alkalis $\rightarrow$ depends on degree of ionisation.

- Strong alkalis fully ionise to produce OH^- ions. (pH 13-14)
- Weak alkalis partially ionise to produce lower concentration of OH^- ions. (pH 8-11)

Strong alkalis $\rightarrow \text{NaOH}, \text{KOH}, \text{Ba(OH)}_2, \text{Ca(OH)}_2$

Weak alkalis $\rightarrow \text{NH}_3(\text{aq})$

Properties of Alkalis

1. Physical Properties of Alkalis

- Alkalis have a bitter taste.
- Alkalis feel slippery & soapy.
- Alkalis can conduct electricity. (presence of mobile ions)
- Alkalis turn red litmus paper blue.
- Alkalis turn Universal indicator blue or violet.
- Alkalis have pH greater than 7.

2. Chemical Properties of Alkalis

- Alkalis react with acids to form salt and water. (neutralisation)

Chemical Reaction: $\text{alkali} + \text{acid} \rightarrow \text{salt} + \text{water}$

- Alkalis react with ammonium salts on heating to form a salt, water and ammonia gas.

Chemical Reaction: $\text{alkali} + \text{ammonium salt} \rightarrow \text{salt} + \text{water} + \text{ammonia gas}$

Determine the salt produced:

positive ion of alkali + negative ion of ammonium

Test for ammonia gas

- Insert a moist red litmus paper at the mouth of the test tube.
- Ammonia gas turns moist red litmus paper blue.

★ IONIC EQUATION: $\text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{NH}_3(\text{g}) + \text{H}_2\text{O}(\text{l})$

pH

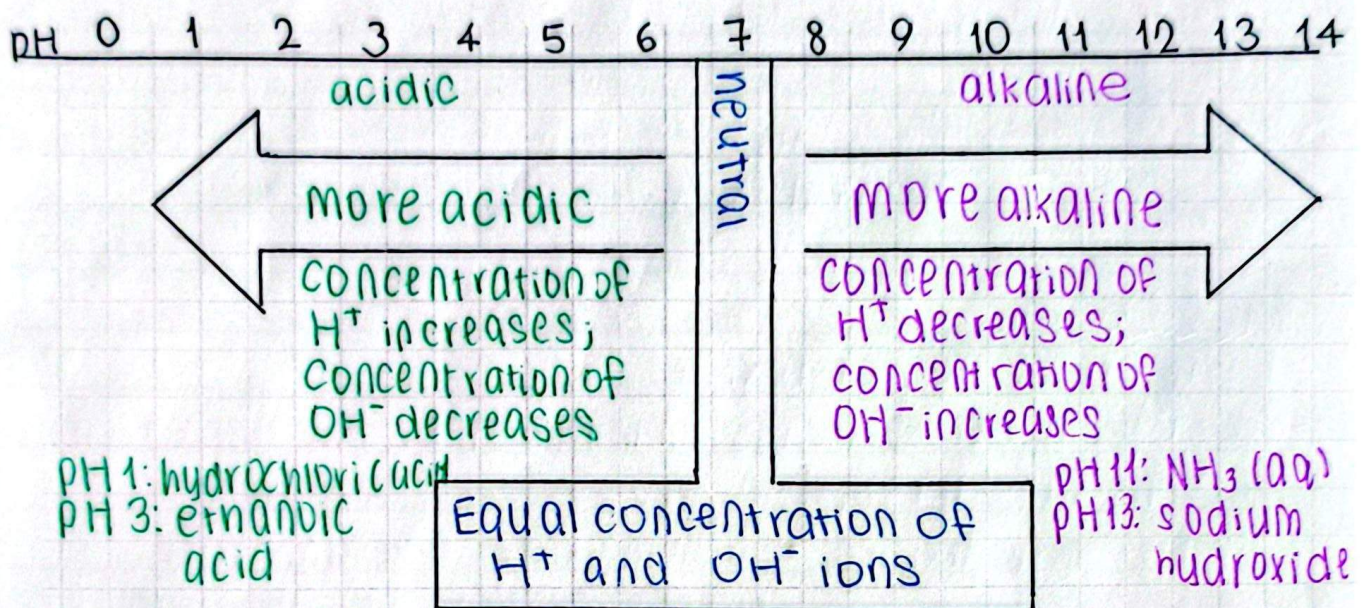
pH is the measure of acidity or alkalinity of a solution.

Acids $\rightarrow \text{pH} < 7 \rightarrow [\text{H}^+] > [\text{OH}^-]$

Neutral solutions $\rightarrow \text{pH} = 7 \rightarrow [\text{H}^+] = [\text{OH}^-]$

Alkalis $\rightarrow \text{pH} > 7 \rightarrow [\text{H}^+] < [\text{OH}^-]$

pH scale



Universal Indicator

pH	colour	nature of solution
0-2	red	strong acid
3-4	orange	weak acid
5-6	yellow	weak acid
7	green	neutral
8-11	blue	weak alkali
12-14	violet	strong alkali

Controlling the pH of soils

Most plants grow best in neutral or slightly acidic conditions.

Treating acidic soils

• Quicklime (calcium oxide)

• Slaked lime (solid calcium hydroxide)

→ increases pH of soil

Oxides (anything with oxygen)

When a metal or non-metal reacts with oxygen, an oxide is formed.

→ metal oxide

Types of oxides: Basic oxide, Amphoteric oxide, Acidic oxide, Neutral oxide → non-metal oxide

Basic oxides

- Basic oxides are metal oxides.
- Most basic oxides are insoluble in water while those that dissolve in water form alkalis.
- Soluble basic oxides → sodium oxide, calcium oxide, potassium oxide, barium oxide
- Basic oxides react with acids to form salt and water.

Amphoteric oxides

- Amphoteric oxides are metal oxides.
- Amphoteric oxides are insoluble in water.
- Amphoteric oxides → lead (II) oxide, aluminium oxide, zinc oxide
- Amphoteric oxides react with both acids and bases to form a salt and water.

(Behave as acids when reacted with bases and behave as bases when reacted with acids.)

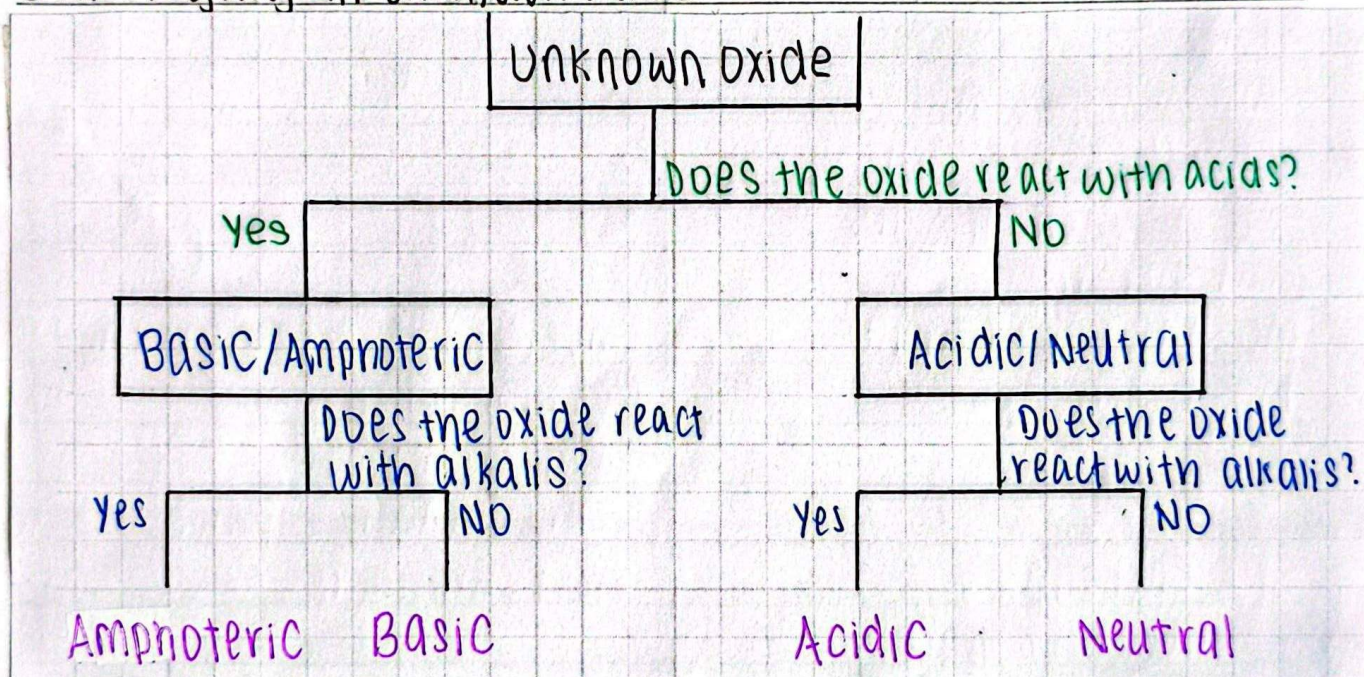
Acidic oxides

- Acidic oxides are non-metal oxides.
- Acidic oxides that are soluble in water dissolve to form acids. (most acidic oxides are soluble)
- Acidic oxides react with bases to form salt and water.

Neutral oxides

- Neutral oxides are non-metal oxides.
- Neutral oxides are insoluble in water.
- Neutral oxides $\rightarrow$ water, carbon monoxide, nitric oxide (NO), nitrous oxide (N_2O).
- Neutral oxides exhibit neither acidic or basic properties.
- Neutral oxides do not react with acids and bases.

Classifying an unknown oxide



Chapter 9: salts

salt

A salt is an ionic compound that consists of a cation and an anion.

- Cation → metal / ammonium ion

- Anion → negative ion of acid

Solubility of salts in water

Soluble salts	Insoluble salts
SPAN	
- All Sodium salts	
- All Potassium salts	
- All Ammonium salts	
- All Nitrates	
MOST Halides (Grp 17)	Silver Halides & Lead(II) Halides
Most Sulfates	Barium Sulfate
	Lead(II) sulfate
	Calcium sulfate → sparingly soluble
SPA Carbonates	Most Carbonates
- Sodium carbonate	
- Potassium carbonate	
- Ammonium carbonate	

Methods to preparing a salt

1. Reaction of Acid with ^{EXCESS} Insoluble substance

- One reactant soluble in water, one reactant insoluble in water
- Salt soluble in water

2. Titration

- Both reactants soluble in water
- Salt soluble in water

3. Precipitation

- Both reactants soluble in water
- Salt insoluble in water

Reaction of Acid with Excess Insoluble substance

Possible Reactions

- Acid + Insoluble Base
- Acid + Metal
- Acid + Insoluble Carbonate

Procedure

Step 1: Add insoluble substance to acid in a beaker until no more insoluble substance reacts. Observe excess unreacted insoluble substance at bottom of beaker.

→ Ensures that all acid has reacted with insoluble substance. Otherwise salt solution will be contaminated with acid.

Step 2: Heat the mixture to increase speed of reaction.

Step 3: Filter the mixture with a filter funnel and filter paper to obtain the filtrate, (salt solution) and to remove excess unreacted insoluble substance.

Step 4: Heat the filtrate to saturation and cool the saturated solution to allow for crystallisation and salt crystals to form.

→ Crystallisation is used so the salt does not thermally decompose and lose its water of crystallisation.

Step 5: Filter the salt solution to obtain crystals.

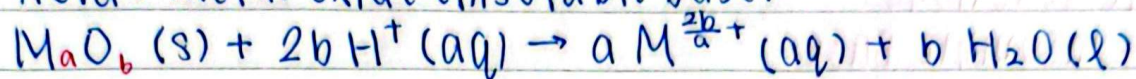
Rinse the (salt) crystals with a few drops of cold deionised water to remove impurities.

→ Little water is used as too much water causes salt crystals to dissolve.

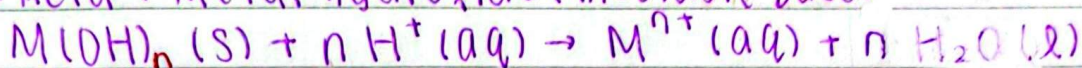
Step 6: Dry crystals by pressing crystals between sheets of filter paper.

Ionic Equation: Reaction of Acid with Insoluble substance

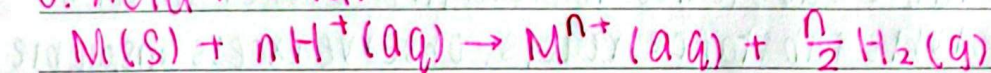
1. Acid + Metal oxide (Insoluble Base)



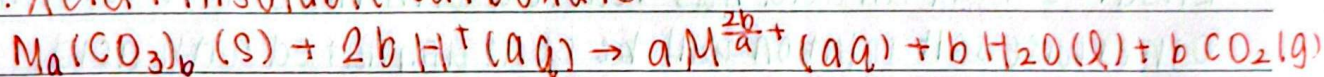
2. Acid + Metal hydroxide (Insoluble Base)



3. Acid + Metal



4. Acid + Insoluble carbonate



Titration

Possible Reactions

- Acid + Soluble Base (SPA Hydroxide)

- Acid + Soluble Carbonate (SPA Carbonate)

OR when salt produced is SPA salt.

Procedure

Step 1: Pipette 25.0 cm^3 of acid into a conical flask. [Analyte]
Add a few drops of indicator into the conical flask.
Colourless (acid) turns (colour).

Step 2: Fill a burette with SPA Hydroxide / carbonate. [titrant]
Note the initial reading on burette (V_i).

Step 3: Titrate the SPA Hydroxide / carbonate into the conical flask ^{while swirling the conical flask} until indicator turns from (colour) to (colour). This signals the end-point of titration.

Step 4: Record the final burette reading (V_2)
Volume of SPA Hydroxide/ carbonate required for complete neutralisation = $(V_2 - V_1) \text{ cm}^3$.

Step 5: Repeat the titration without adding indicator to obtain pure salt solution.

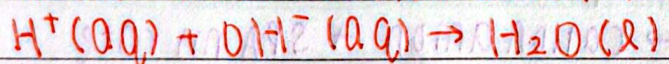
Step 6: Heat the salt solution to saturation and cool the saturated solution to allow for crystallisation and salt crystals to form.

Step 7: Filter the salt solution to obtain crystals. Rinse the (salt) crystals with a few drops of cold deionised water to remove impurities.

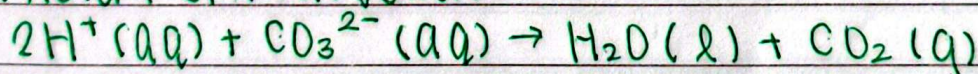
Step 8: Dry crystals by pressing crystals between sheets of filter paper.

IONIC Equation: Titration

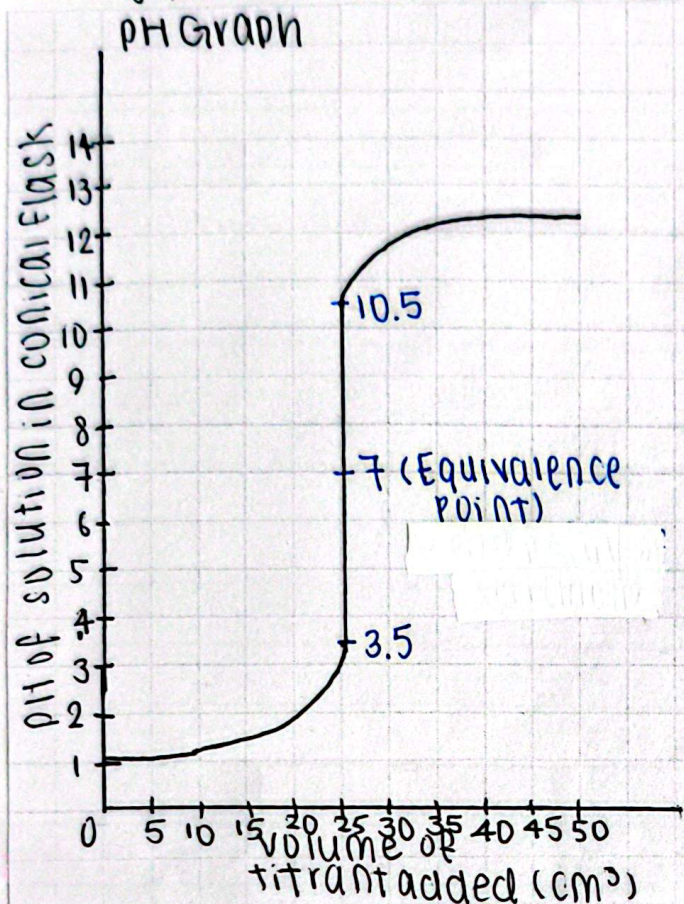
1. Acid + Alkali (SPA Hydroxide)



2. Acid + SPA Carbonate



Using pH meter to detect end-point of titration



Equivalence point

→ Occurs when moles of

acid = moles of base

→ Find top and bottom point of vertical line.

Take average of the 2

points to get equivalence

point.

Precipitation

To obtain an insoluble salt (cation anion),

cation nitrate + SPA anion → cation anion + SPA nitrate
(insoluble salt)

Procedure

Step 1: Mix cation nitrate solution and SPA anion solution together in a beaker.

Stir the mixture to allow the insoluble cation anion to form (precipitate).

Step 2: Filter the mixture using a filter funnel and filter paper and collect the insoluble salt as residue.

Step 3: Rinse the insoluble salt with cold deionised water to remove impurities.

Step 4: Dry the insoluble salt between sheets of filter paper.

Ionic Equation: Precipitation

